OriginPass

El origen no se imita.

Executive infographic · Deliverable 4

ST0251 Proyecto Integrador 1, 2026-2

Universidad EAFIT · Medellín, Colombia

Prove where a product came from, without having to trust the seller.

PROJECT OVERVIEW

OriginPass — a verifiable digital passport for every physical product.

Team: Emmanuel Hernández Melo, individual project.

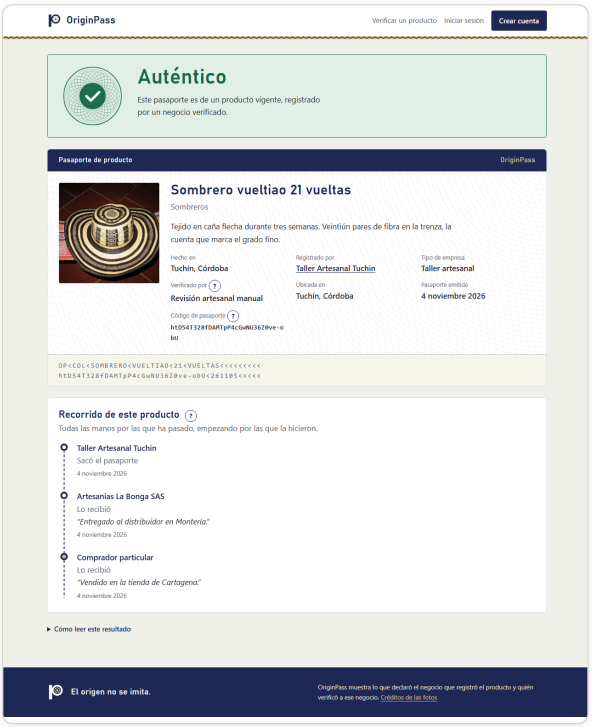
Product Owner: Emmanuel Hernández Melo, who also acts as Scrum Master and developer.

PROBLEM

Imitations of Colombian crafts and small brands are sold next to the genuine piece. Printed certificates can be photocopied, brand-owned apps serve only large brands, and a buyer at a market stall cannot check, before paying, who made a piece or how it reached the stall.

PROPOSED SOLUTION

OriginPass issues a digital passport for every physical unit, with a QR code that anyone can scan without an account or an app. The public page gives a verdict — genuine, revoked, altered or not found — names the verified producer and shows the chain of custody from the maker to the current owner. Companies are approved through an official registry and artisan workshops through a manual review. Every handover is signed and chained, every withdrawal states its reason, and scan analytics can warn a producer when one code appears in two regions at once.



System snapshot, taken from the deployed application: the verdict Auténtico, the passport and its chain of custody, on a computer and on a 360 px phone.

IMPLEMENTED FEATURES

Verify with no account
Scan the QR and read the verdict: genuine, revoked, altered or not found. FR28–FR31, FR62

One passport per unit
A code that cannot be derived from another, and a printable QR with the OriginPass mark. FR21, FR22, FR61

Chain of custody
Every handover is signed and chained; the buyer claims the piece with two codes, reading the QR by camera or from a photo. FR32, FR35–FR40, FR60

Verified producers
Official registry for companies, manual review for artisan workshops, and a public profile. FR07–FR18

Revocation and audit
A withdrawal always states its reason, and every status change goes to a signed audit log. FR41–FR45

Scan analytics
Scans by date and region, product ranking, trend, CSV export and a duplicate-scan alert. FR46–FR52

Accounts and abuse resistance
Registration, login lockout, password reset and change, and a cap on stored scans. FR01–FR06, FR57–FR59

54 / 62 functional requirements delivered, 5 more in part

349 automated tests passing on PostgreSQL

HTTPS deployed on AWS EC2, Let's Encrypt certificate

TECHNOLOGY STACK

Python · Django 5.2 · PostgreSQL · gunicorn · nginx · WhiteNoise · AWS EC2 · Ubuntu 26.04 · Let's Encrypt · qrcode + Pillow · jsQR · GitHub Actions · draw.io (UML)

PROJECT LINKS

Source code: github.com/Pacha-e/OriginPass

Wiki: github.com/Pacha-e/OriginPass/wiki

Deployed application: https://13-220-96-49.sslip.io